

NANYANG TECHNOLOGICAL UNIVERSITY

School of Social Sciences

HE3002 Macroeconomics III

PROBLEM SET 7

The IS-MPR-PC Model

1. The IS curve is given by:

$$y_t = y_t^* - \alpha(i_t - \pi_t - r^*) + \epsilon_t^y \quad (1)$$

and the monetary policy rule is given by:

$$i_t = r^* + \pi^* + \beta_\pi(\pi_t - \pi^*) \quad (2)$$

- (a) Derive the slope of the IS-MPR curve and explain why β_π should be larger than 1.
 - (b) Discuss the factors that affect the position of the IS-MPR curve.
2. **Answer parts (a) and (b) for PS 7 and keep (c) for PS 8** Consider an economy in long-run equilibrium. Suddenly, consumer confidence increases significantly, leading to a higher level of consumer spending.
- (a) Illustrate the short-run effect of this change on the IS-MPR-PC model, showing how it impacts output and the price level.
 - (b) Explain what type of output gap is created and why.
 - (c) Describe how the economy might adjust back to long-run equilibrium over time if the central bank does not intervene.
3. Assume an economy experiences a negative supply shock due to a sudden increase in the price of a key input, such as oil.
- (a) Show and explain the short-run effects of this shock on the IS-MPR-PC model, highlighting changes in the aggregate supply curve, output, and price level.
 - (b) If the government aims to stabilize output, describe the fiscal or monetary policy tools it might use. Discuss any potential trade-offs associated with these policies in the short run.
4. **Answer part (a) and keep (b) and (c) for PS 8** The central bank in a country decides to reduce the interest rate in an attempt to boost the economy. Assume the economy is initially in long-run equilibrium.
- (a) Using the IS-MPR-PC model, describe the short-run effects of the interest rate reduction on aggregate demand, output, and the price level.

- (b) Explain how the economy might adjust over time if the central bank keeps the interest rate low. What are the potential long-run impacts on output and the price level?
- (c) Discuss whether a reduction in the interest rate can have a permanent effect on real output in the IS-MPR-PC model framework, and explain why or why not.